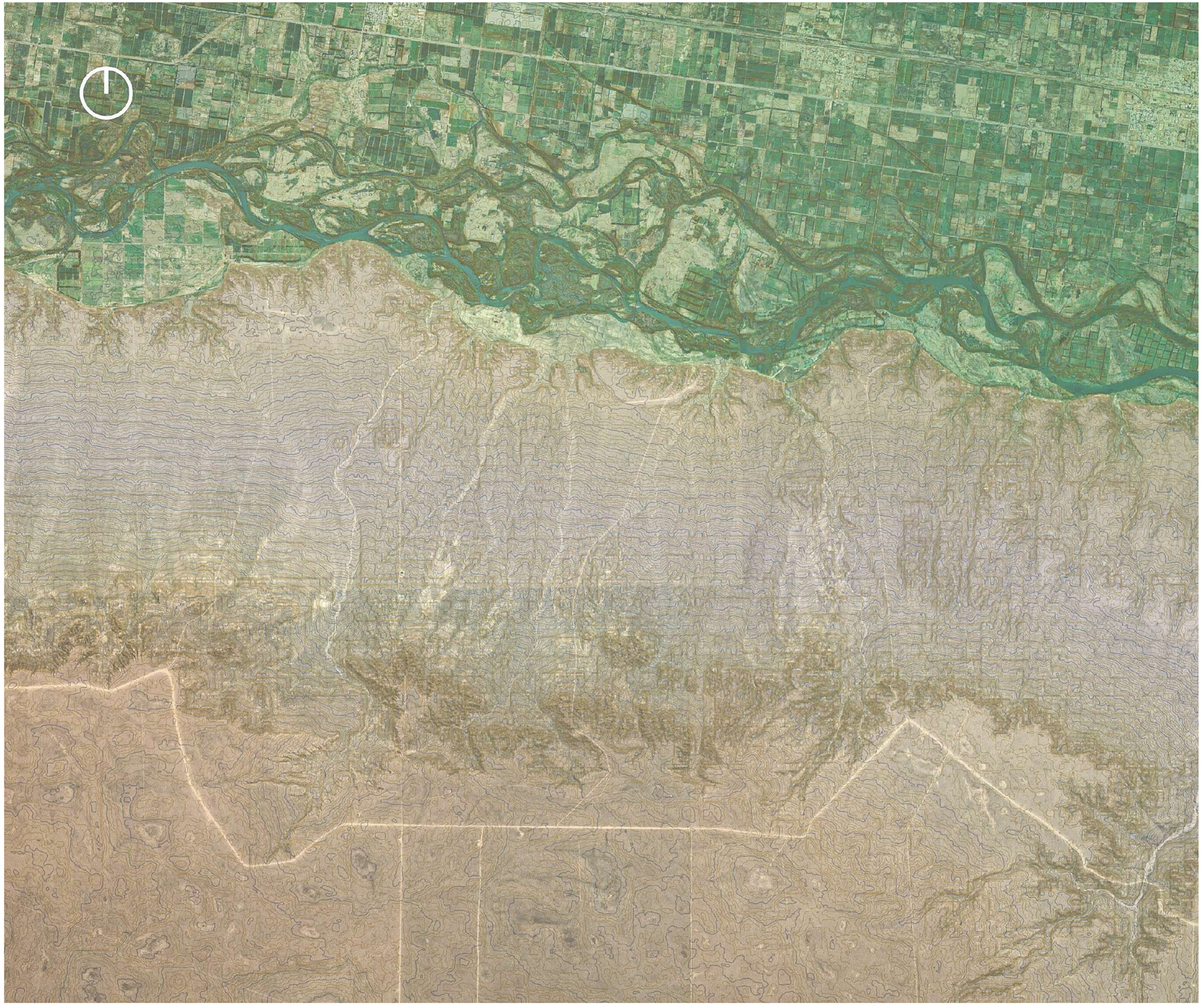


Lighthouse Patagonia

A beacon for the new era of regenerative developments.

Valle de Río Negro



Manifesto

A light at the edge of the desert.

A place that shows how water, land and people can grow life together.

The problem

Fresh water is becoming the defining resource of this century. Climate change is making droughts longer and storms more violent. Cities keep expanding over the very land that feeds them. In arid regions, the little rain that falls runs off before the ground can hold it, and it carries the soil away with it.

What we believe

Human intervention can do more than limit damage. With careful design, land can be helped to hold water, build soil and host more life than it did before. On the river plain, that means restoring the riparian forest. Beyond the barda, it means something more ambitious: slowly winning living ground from the desert.

Our answer

LightHouse VRN is organized as a Regenerative Agroecological Village. It is a low-density, rururban settlement that integrates bioclimatic homes, diversified agro-ecological production and the active conservation of the riparian forest and the barda. It is not a conventional subdivision of identical plots cut off from the land's productive role. It is a place where living on the land and caring for it are the same act.

The edge: the Río Negro and its valley meet the barda and the Patagonian plateau. Mainqué, Río Negro, Argentina.



North Patagonia River System

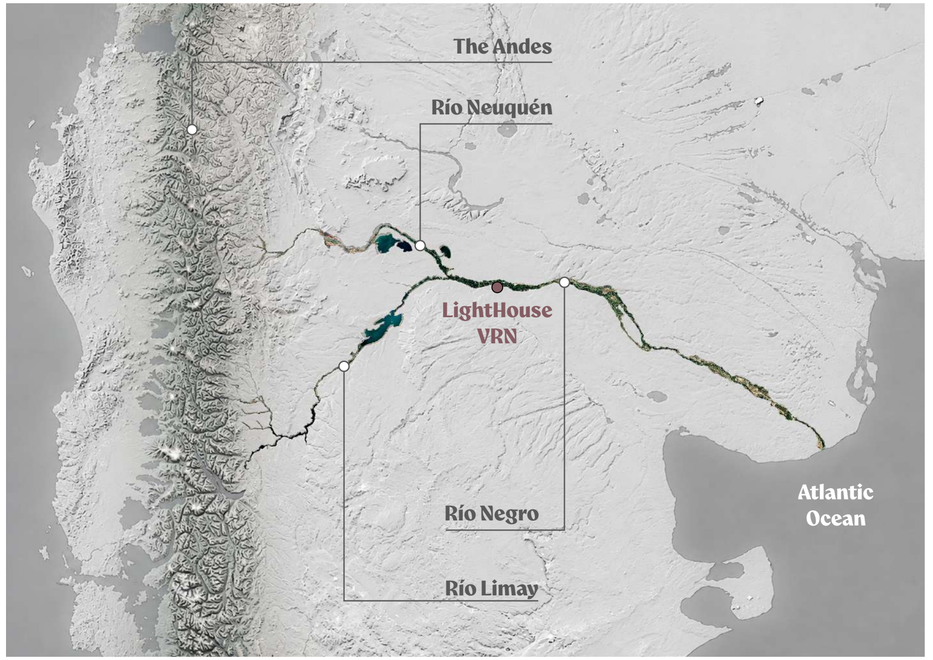
The network from the Andes to the Atlantic Ocean.

High in the Andes, rain and snow fall at up to 3,000 mm a year. From there, two rivers descend toward the steppe: the Limay, born in the lakes of the south, and the Neuquén, from the northern peaks.

They meet at the city of Neuquén and become the Río Negro. The Limay brings about 70% of the water, the Neuquén the remaining 30%. From there, the river crosses 640 km of semi-arid land, where barely 200 mm of rain falls, all the way to the Atlantic, without receiving a single tributary. Together they drain 140,000 km², the most important river system lying entirely within Argentina.

Every second, the Río Negro carries an average of 930 m³ of water: around 29 billion m³ a year, about 1.6 times the natural flow of the Colorado River in the United States. A chain of reservoirs upstream regulates that flow, softening floods and droughts and storing more than a full year of it. And only a small share is used: around 11 to 16% of the river's average flow is withdrawn today for farms, towns and industry.

Sources: Autoridad Interjurisdiccional de las Cuencas de los Ríos Limay, Neuquén y Negro (AIC); FAO UTF/ARG/017; Río Negro Water Master Plan (CFI, 2024); ORSEP; US Bureau of Reclamation.

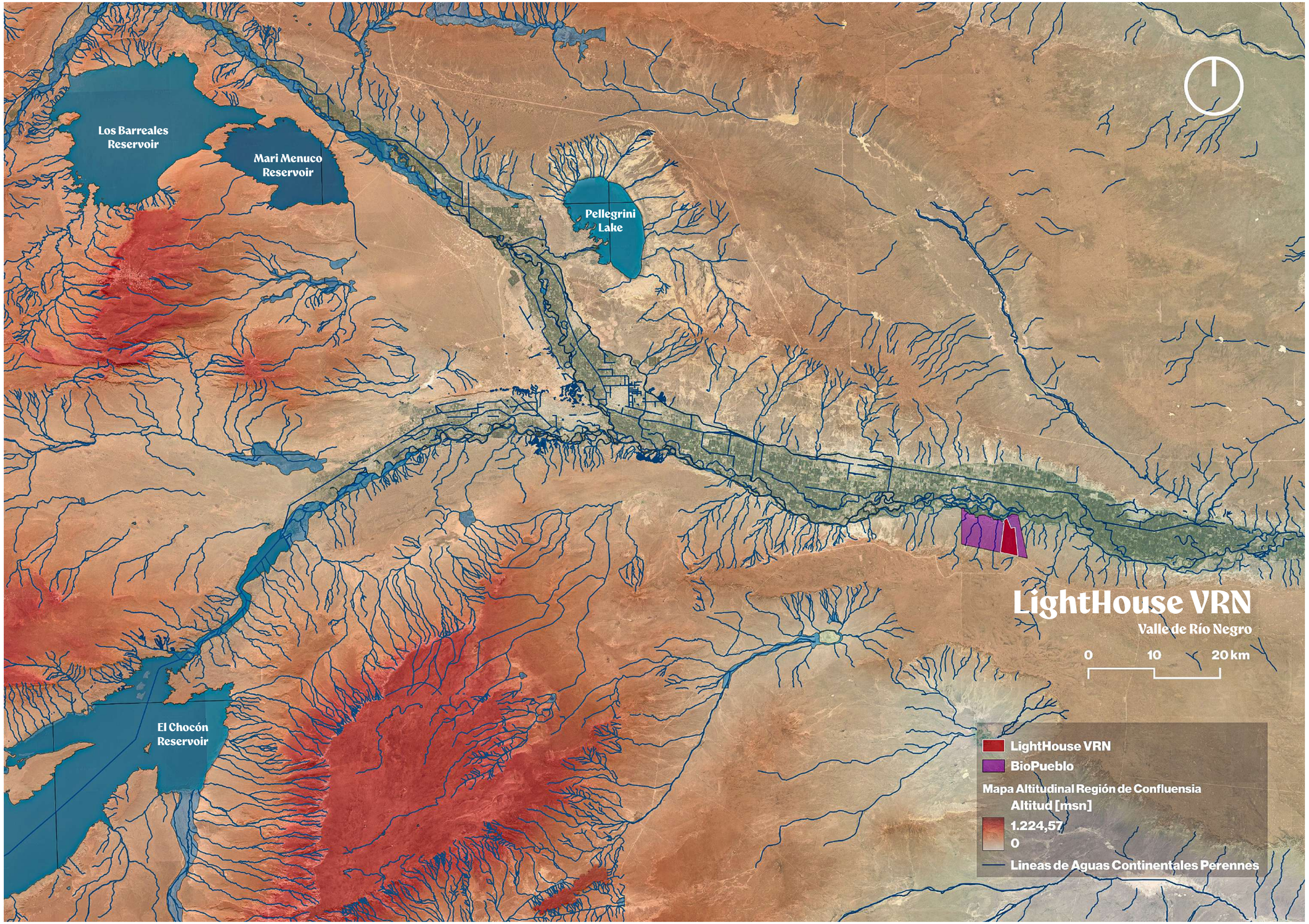












A State of Abundance

Water. Energy. Food.

One river, three forms of abundance.

At Neuquén–Cipolletti, Río Limay (70% of the flow) and Río Neuquén (30%) meet to form the RíoNegro.

Water

Upstream, a chain of reservoirs holds more than a full year of the river's flow, the largest stored-water surface in Argentina.

Energy

The same dams produce a quarter of the country's hydropower: 4.6 GW of clean, renewable electricity.

Food

Downstream, a century-old network of canals turns desert into the Upper Valley, a green strip of orchards that grows most of Argentina's pears and makes it the world's second-largest pear exporter.

LightHouse VRN sits at the edge of this oasis, where the valley meets the desert.

Sources: AIC; Instituto Geográfico Nacional / Energía Argentina (CAMMESA data, 2022); ORSEP; Government of Río Negro.

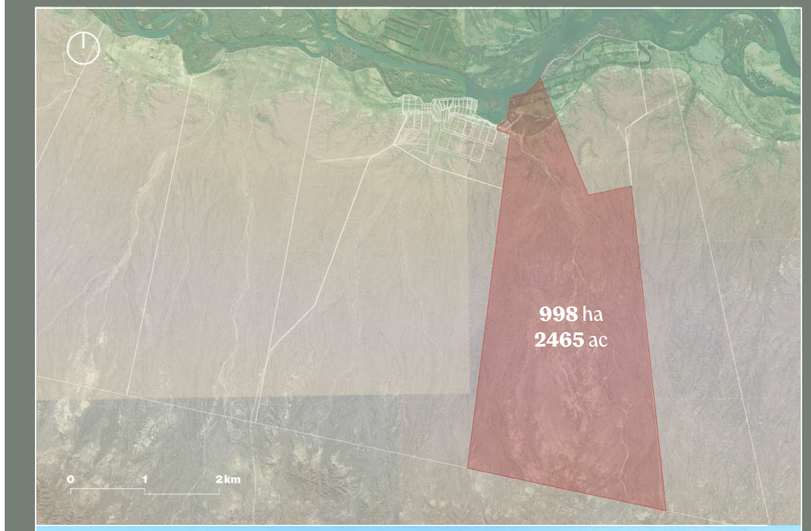
Land Analisis & Research

LightHouse
Valle de Río Negro

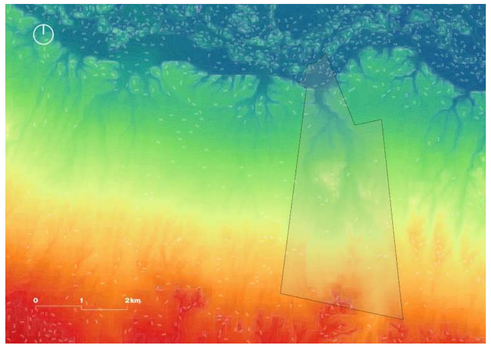
A story written by Water & Desert.

LightHouse VRN occupies nearly 1,000 hectares on the south bank of the Río Negro, at Mainqué. In a single property, the land runs from the river's channels and islands, through an alluvial floodplain of riparian forest, up the barda escarpment and onto the semi-arid plateau.

Across the plateau, dry washes carry the rain of rare, violent storms down toward the valley. That rain rushes through and is lost, taking the soil with it.







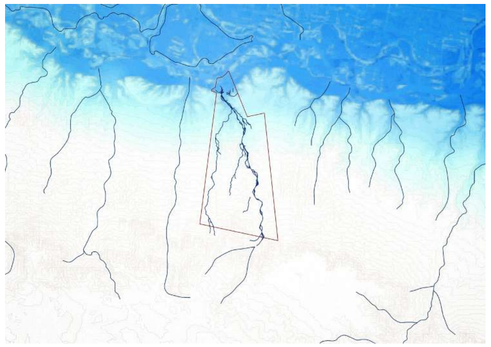
Topographic Map

A fully serviced, titled property within a living ecosystem, located in **one of the fastest-growing areas** of Patagonia. Three **principal houses**, two **cabins**, **roads**, **power**, **fiber**, an EANA-authorized **helipad**,



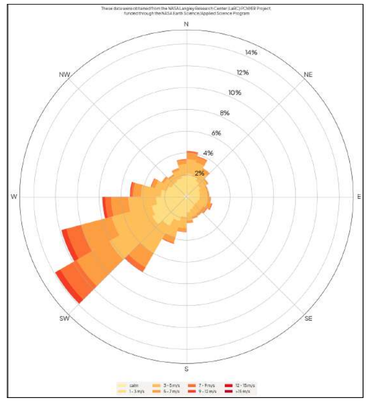
Hillshade Map

A fully serviced, titled property within a living ecosystem, located in **one of the fastest-growing areas** of Patagonia. Three **principal houses**, two **cabins**, **roads**, **power**, **fiber**, an EANA-authorized **helipad**,



Hydrological & Runoff Map

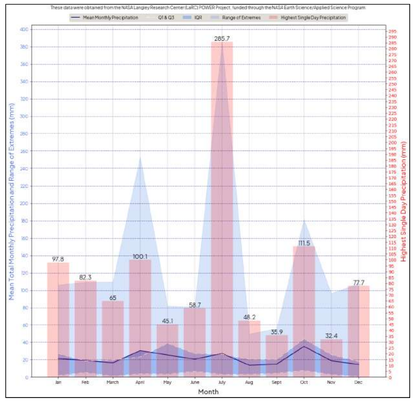
A fully serviced, titled property within a living ecosystem, located in **one of the fastest-growing areas** of Patagonia. Three **principal houses**, two **cabins**, **roads**, **power**, **fiber**, an EANA-authorized **helipad**,



Annual Wind Patterns

Annual wind rose showing the dominant wind directions and frequencies for the site.

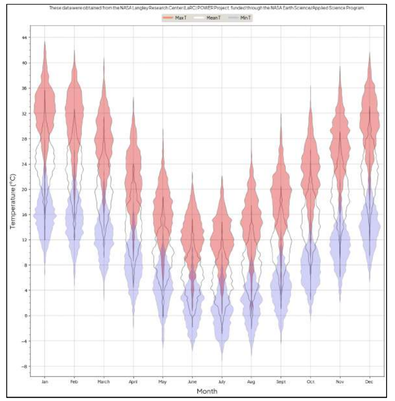
This supports general site orientation and the placement of structures, vegetation, and sensitive infrastructure.



Precipitation

Monthly precipitation summary. The navy blue line represents average monthly totals, with shaded bands indicating variability: the interquartile range (darker blue) and full observed range (lighter blue).

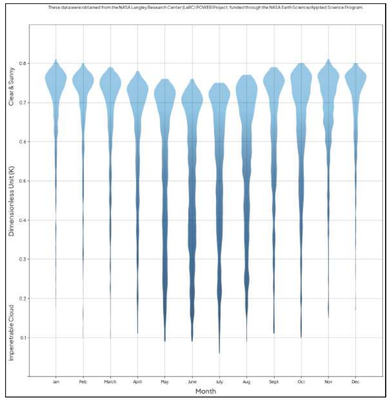
Red bars show the highest single-day rainfall recorded in each month.



Temperature

Monthly distribution of daily mean (white), maximum (red), and minimum (blue) temperatures. Violin plot widths reflect the frequency of values, while the vertical range represents observed extremes.

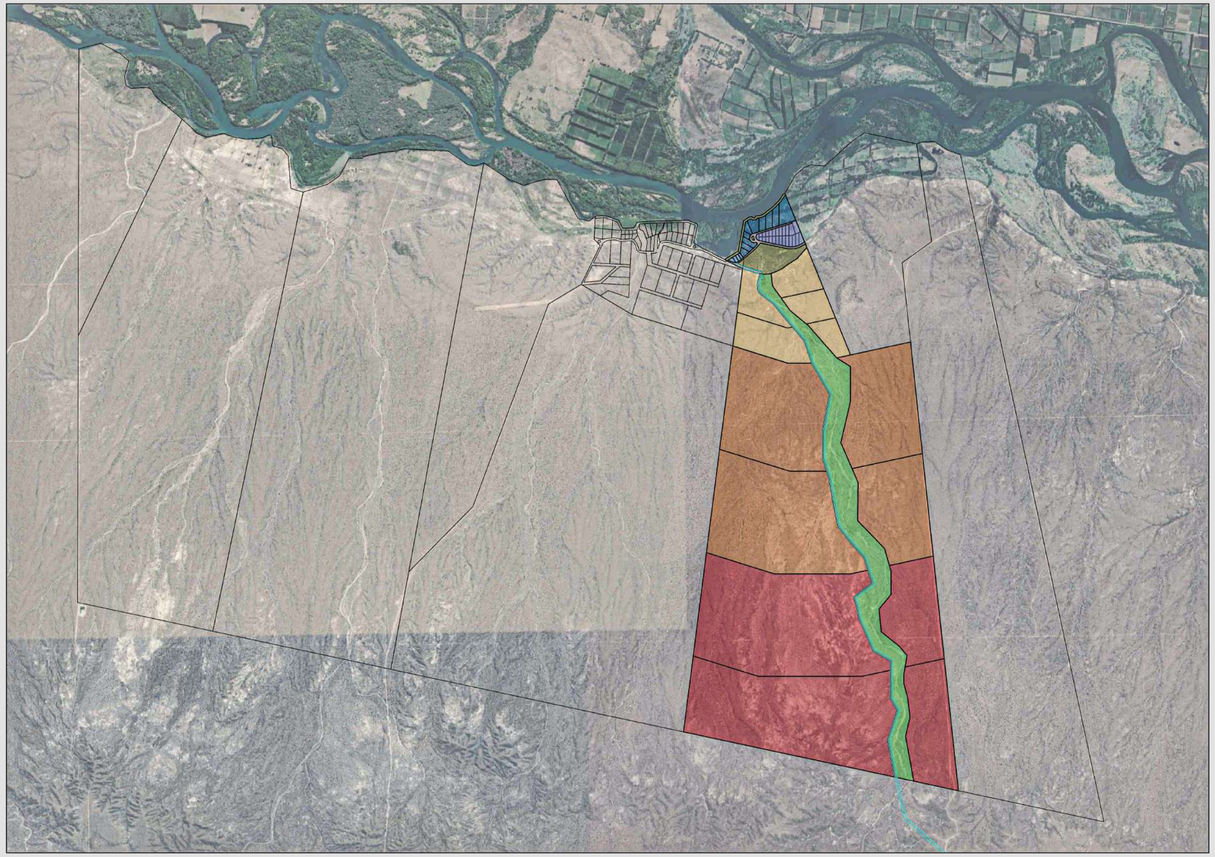
This visualization captures both typical and exceptional temperature conditions throughout the year.



Solar Insolation

Monthly distribution of daily clear-sky solar insolation index values over 30 years. Each violin plot shows the frequency of daily sky clearness values for that month, with wider sections representing more frequent conditions. A color gradient from dark (low clearness, "Impenetrable Cloud") to light (high clearness, "Clear & Sunny") illustrates the scale. Note: this index reflects relative sky clarity and does not represent actual solar energy flux.

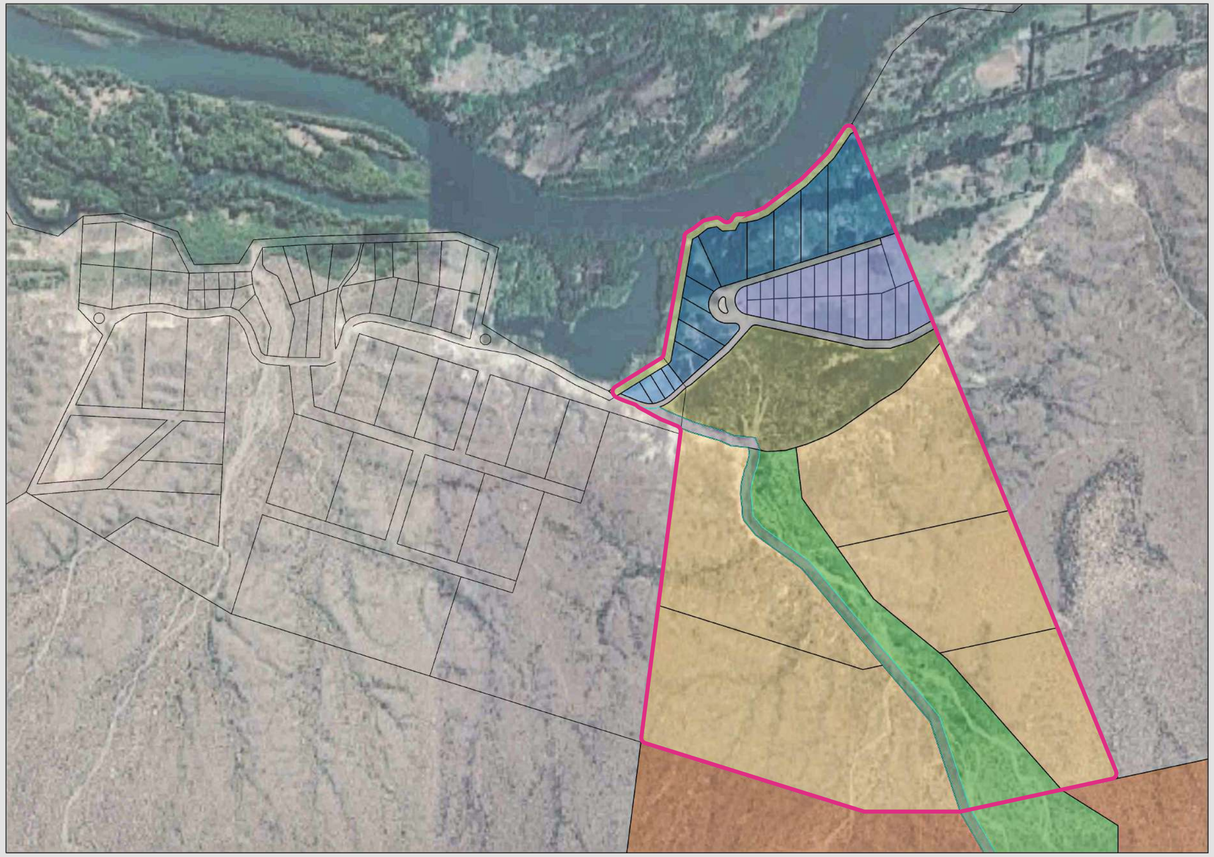
V-01
Lineamientos Regenerativos & Subidivisión Preliminar



Cañadón principal: eje estructurante del proyecto. Se preserva el cauce natural de escurrimiento pluvial y se lo convierte en la columna vertebral del desarrollo regenerativo además de cumplis con el requerimiento normativo de espacios verdes. Su capacidad para retener e infiltrar el agua de lluvia, junto con la sombra y el reparo del viento que ofrece, lo convierten en el ámbito más propicio para establecer especies perennes nativas y adaptadas.

Nuevo camino principal Norte-Sur (30m de ancho según normativa) que conecta la costa ribereña de LightHouse y BioPueblo con la Ruta Provincial 7, brindando acceso independiente al proyecto y cociendo verticalmente todas sus etapas y lotes.

Etapa 01 | 118ha



Camino de Sirga: respetando normativa y cota de inundación | Ancho:20m

Lotes Ribereños Etapa 0 | Cantidad: 4 | Superficie: 7316m² - 0,73ha

Lotes Ribereños Etapa 1 | Cantidad: 10 | Superficie: 10,10ha

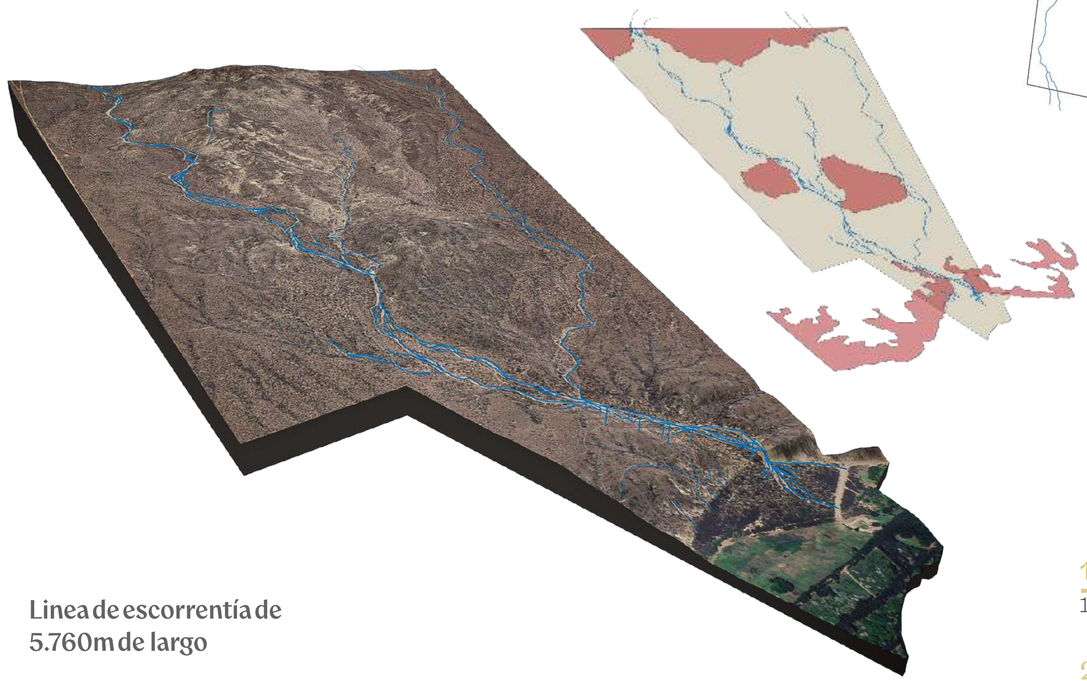
Lotes Residenciales Etapa 1 | Cantidad: 24 | Superficie: 7,10ha

Área de Preservación y Restauración del Ecosistema Bosque Ribereño | Superficie: 11,3ha

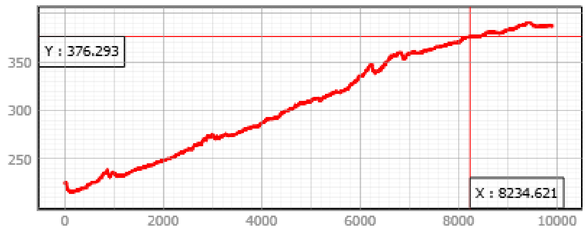
Lotes de Uso Agrícola | Cantidad: 5+ | Superficie: 69,7ha

Área Natural Protegida + Camino Principal | Superficie: 13,9ha - Requerido por Normativa: 11,8ha

Water is the asset.
Regeneration is the strategy.
The village is the outcome.



Linea de escorrentía de
5.760m de largo

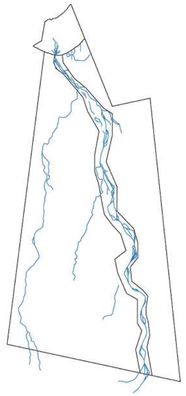


Casi 1000ha con los nuevos lotes.

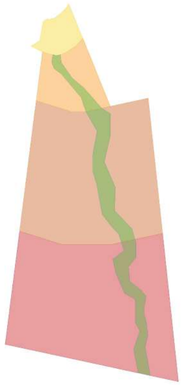
10% deben ser areas verdes: 100ha

5% deben ser reservas fiscales: 50ha

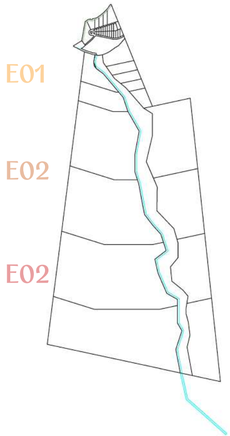
Escorrentía



Zonificación



Etapas



Zona A
Regeneration Project
35ha

Planicie Aluvial con
Bosque Ribereño

Zona B
Nature-positive Project
83,5ha

Barda Baja

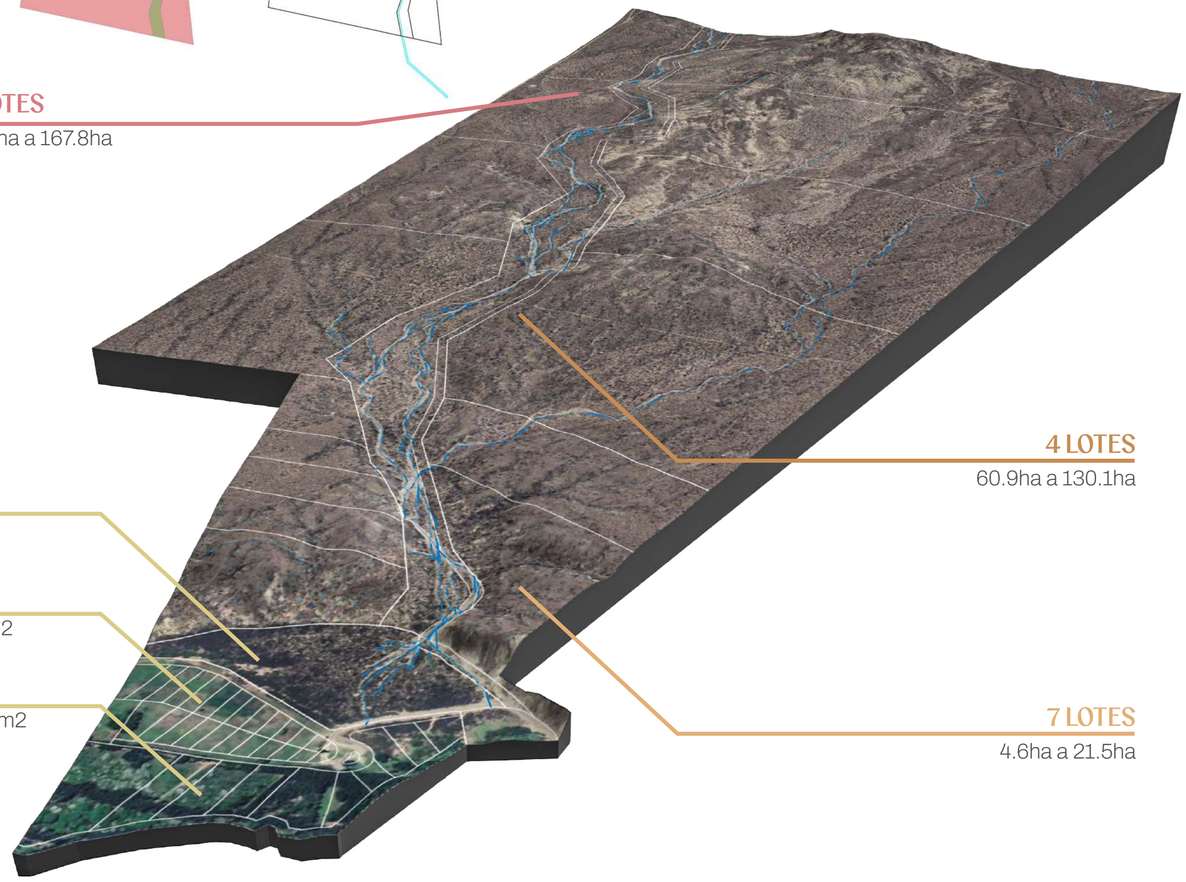
Zona C
Nature-positive Project
414,7ha

Barda Alta

Zona D
Nature-positive Project
464,8ha

Corniza Meceta Patagónica

4 LOTES
50.6ha a 167.8ha



4 LOTES
60.9ha a 130.1ha

1 LOTE
1.13ha

24 LOTES
1.200m2 a 5.800m2

10 LOTES
4.900m2 a 29.800m2

7 LOTES
4.6ha a 21.5ha